

**Textbooks and Resources:**

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**ATI Book:**

Assessment Technologies Institute (ATI). (2023). RN adult medical-surgical nursing (Edition 12.0) Assessment Technologies Institute / Ascend Learning.

**ATI Online Resources:**

Assessment Technologies Institute (ATI). (2026). Assessment Technologies Institute.  
[www.atitesting.com](http://www.atitesting.com)

**Lippincott Textbook:**

Hinkle, J. L., Cheever, K. H., & Overbaugh, K. (2022). *Brunner & Suddarth's textbook of medical-surgical nursing* (15<sup>th</sup> ed.). Wolters Kluwer Health.

Textbook ISBN Numbers: 978-1-9751-6103-3 & 978-1-9751-6103-3

Note: chapters with \* are secondary chapters that support the primary.

**Course Content - Maximizing Your Learning: Course Materials & ATI Modules.**

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To succeed in this course, students are encouraged to review assigned readings and course materials thoroughly before each lecture including those in the Canvas course shell. Active engagement with the content before class will make lecture time more meaningful and allow for deeper understanding.

Completion of ATI modules is strongly recommended. These modules are designed to reinforce and solidify key concepts, helping you connect theory to practice. Plan to dedicate approximately 2 hours of study for every 1 hour of lecture. Consistent, focused study time is one of the strongest predictors of success in nursing courses.

When reviewing chapters, pay special attention to exemplars and examples, which are included under the chapter subheadings. Using these as a guide will help you locate relevant information quickly and connect examples to broader concepts.

Learning objectives are designed to align with the recommended concept-mapping framework used throughout the course – emphasizing pathophysiology, etiology, assessment findings, diagnostics, nursing interventions, and treatments.

★ Remember: Nursing school is challenging, but it's also incredibly rewarding. Staying organized, reviewing materials consistently, and actively practicing concepts through ATI modules will help you build confidence and mastery. Take advantage of these resources, and don't hesitate to reach out if you need guidance. Your effort now will pay off in your clinical and professional growth!!

**EXEMPLARS ORGANIZED BY LECTURE DAY**

***Lecture Day 1: Electrolyte Balance Concepts***

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**Concept focus:** balance of electrolytes across body systems and their effects on cellular, neuromuscular, and cardiac function.

**Objectives:**

By the end of this learning experience, the student will be able to:

1. **Describe** the role of major electrolytes in maintaining cellular, neuromuscular, and cardiac function. (QSEN: Patient-Centered Care, Safety)
2. **Identify** assessment findings and diagnostic data associated with common electrolyte imbalances. (QSEN: Informatics, Safety)
3. **Explain** the clinical manifestations of sodium, potassium, calcium, magnesium, phosphorus, and chloride imbalances. (QSEN: Patient-Centered Care, Safety)
4. **Apply** appropriate nursing interventions for patients experiencing electrolyte imbalances across the lifespan. (QSEN: Patient-Centered Care, Teamwork & Collaboration, Safety)
5. **Analyze** patient responses to electrolyte disturbances and related treatments to support clinical judgment. (QSEN: Evidence-Based Practice, Safety)

**Exemplars:**

- Sodium Imbalances (Deficit & Excess)
- Potassium Imbalances (Deficit & Excess)
- Calcium Imbalances (Deficit & Excess)
- Magnesium Imbalances (Deficit & Excess)
- Phosphorus Imbalances (Deficit & Excess)
- Chloride Imbalances (Deficit & Excess)

**Review:**

**ATI:**

*Book: Chapter 45, Page 309*

*ATI Online Resources: Engage Adult Medical Surgical RN > Alterations in Health > Disrupted Homeostasis*

**Lippincott:**

*Chapter 10: Fluid & Electrolytes*

*Chapter 48: Management of patients with Kidney Disorders (electrolyte imbalances)*

**Lecture Day 2: Acid-Base Balance Concepts**

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**Concept focus:** acid-base imbalances and the use of arterial blood gases to understand physiological disruption and clinical implications.

**Objectives:**

By the end of this learning experience, the student will be able to:

1. **Describe** normal acid-base balance and the role of respiratory and renal systems in maintaining physiologic pH. (QSEN: Patient-Centered Care, Safety)
2. **Identify** assessment findings and arterial blood gas values associated with acid-base imbalances. (QSEN: Informatics, Safety)
3. **Differentiate** between metabolic and respiratory acidosis and alkalosis using arterial blood gas interpretation. (QSEN: Informatics, Evidence-Based Practice, Safety)
4. **Apply** appropriate nursing interventions for patients experiencing acid-base imbalances based on clinical findings and ABG results. (QSEN: Patient-Centered Care, Teamwork & Collaboration, Safety)
5. **Analyze** patient responses to acid-base disturbances and related treatments to support clinical judgment. (QSEN: Evidence-Based Practice, Safety)

**Exemplars:**

- Metabolic Acidosis
- Metabolic Alkalosis
- Respiratory Acidosis
- Respiratory Alkalosis

**Review:**

**ATI:**

*Book Chapter 46, Page 319*

*ATI Online Resources: Engage Adult Medical Surgical RN > Alterations in Health > Disrupted Homeostasis*

**Lippincott:**

*Chapter 10: Fluid & Electrolytes*

*Chapter 17: Assessment of Respiratory Function*

*Chapter 48: Management of patients with Kidney Disorders (acid/base imbalances)*

**Lecture Day 3 & 4: Metabolism Concepts**

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**Concept focus:** metabolic control through glucose regulation across the lifespan.

**Objectives:**

By the end of this learning experience, the student will be able to:

1. **Describe** normal glucose metabolism and the role of insulin in maintaining blood glucose homeostasis. (QSEN: Patient-Centered Care, Safety)
2. **Identify** assessment findings and diagnostic data associated with altered glucose regulation, including hypoglycemia and hyperglycemia. (QSEN: Informatics, Safety)
3. **Differentiate** between Type 1 and Type 2 diabetes mellitus based on pathophysiology, clinical presentation, and management principles. (QSEN: Patient-Centered Care, Evidence-Based Practice, Safety)
4. **Apply** appropriate nursing interventions for patients experiencing altered glucose regulation across the lifespan. (QSEN: Patient-Centered Care, Teamwork & Collaboration, Safety)
5. **Analyze** patient responses to therapeutic management of diabetes and related complications to support clinical judgment. (QSEN: Evidence-Based Practice, Safety)

**Exemplars:**

- Diabetes Type 1
- Diabetes Type 2

**Review:**

**ATI:**

*Book: Chapter 83 & 84, Pages 627 & 637*

*ATI Online Resources: Engage Adult Medical Surgical RN > Alterations in Health >*

*Alterations in Endocrine Function >*

*Type 1 Diabetes, Type 2 Diabetes, Hypoglycemia, & Hyperglycemia*

**Lippincott:**

*Chapter 45: Assessment of Endocrine Disorders*

*Chapter 46: Assessment and Management of Patients with Diabetes*

*Chapter 10: Fluid & Electrolytes*

*Chapter 12: Nutrition*

**Lecture Day 5: Cellular Regulation Concepts (Anemias)**

**Concept focus:** alterations in red blood cell production and function and their impact on oxygen delivery to tissues.

**Objectives:**

By the end of this learning experience, the student will be able to:

1. **Describe** normal red blood cell production and the role of hemoglobin in oxygen transport to tissues. (QSEN: Patient-Centered Care, Safety)
2. **Identify** assessment findings and diagnostic data associated with common anemias, including iron deficiency, megaloblastic, and hemolytic anemias. (QSEN: Informatics, Safety)
3. **Differentiate** between types of anemia based on etiology, pathophysiology, and clinical manifestations. (QSEN: Patient-Centered Care, Evidence-Based Practice, Safety)
4. **Apply** appropriate nursing interventions for patients experiencing anemia across the lifespan. (QSEN: Patient-Centered Care, Teamwork & Collaboration, Safety)
5. **Analyze** patient responses to treatment and management of anemia to support clinical judgment and ongoing care. (QSEN: Evidence-Based Practice, Safety)

**Exemplars:**

- Cellular Regulation
- Iron Deficiency Anemia
- Megaloblastic Anemias
  - Pernicious Anemia
  - Folic Acid Anemia
- Hemolytic Anemia
  - Sickle-Cell Anemia

**Review:**

**ATI:**

*Book: Chapter 42; Page 293*

*ATI Online Resources: Engage Adult Medical Surgical RN > Alterations in Health > Alterations in Hematologic Function*

**Lippincott:**

*Chapter 30: Assessment of Hematologic Function*

*Chapter 31: Management of Patients with Hematologic Disorders*

*\*Chapter 8: Genetic Considerations*

*\*Chapter 12: Nutrition*

**Lecture Day 6: Managing Care Concepts**

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**Concept focus:** nursing management of preoperative preparation and postoperative recovery.

**Objectives:**

By the end of this learning experience, the student will be able to:

1. **Describe** the nurse's role in preparing patients for surgical procedures and supporting recovery following surgery. (QSEN: Patient-Centered Care, Safety)
2. **Identify** assessment findings and diagnostic data relevant to preoperative and postoperative care. (QSEN: Informatics, Safety)
3. **Apply** appropriate nursing interventions to promote patient safety and optimize outcomes during the preoperative and postoperative phases of care. (QSEN: Patient-Centered Care, Teamwork & Collaboration, Safety)
4. **Analyze** patient responses to surgical intervention, including pain, respiratory status, and potential complications. (QSEN: Evidence-Based Practice, Safety)
5. **Prioritize** nursing care for patients in the postoperative period to support recovery and prevent and identify complications. (QSEN: Patient-Centered Care, Safety)

**Exemplars:**

- Preoperative Care
- Postoperative Care

**Review:**

**ATI:**

*Book: Chapter 96 & 97; Pages 751 & 757*

*ATI Online Resources: Engage Adult Medical Surgical RN > Alterations in Health > Caring for the Surgical Client*

**Lippincott:**

*Chapter 14: Preoperative Nursing Management*

*Chapter 16: Postoperative Nursing Management*

*\*Chapter 9: Pain Management*

*\*Chapter 17: Assessment of Respiratory Function*

**Lecture Day 7: Perfusion Concepts**

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**Concept focus:** blood flow *to the heart muscle* – supporting myocardial oxygenation and function.

**Objectives:**

By the end of this learning experience, the student will be able to:

1. **Describe** normal coronary circulation and the factors that affect myocardial oxygen supply and demand. (QSEN: Patient-Centered Care, Safety)
2. **Identify** assessment findings, diagnostic data, and clinical indicators associated with impaired coronary perfusion. (QSEN: Informatics, Safety)
3. **Differentiate** between chronic stable angina, unstable angina, and myocardial infarction based on pathophysiology and clinical presentation. (QSEN: Patient-Centered Care, Evidence-Based Practice, Safety)
4. **Apply** appropriate nursing interventions for patients experiencing acute and chronic alterations in coronary perfusion. (QSEN: Patient-Centered Care, Teamwork & Collaboration, Safety)
5. **Analyze** patient responses to treatment and early recognition of complications associated with acute coronary syndrome. (QSEN: Evidence-Based Practice, Safety)

**Exemplars:**

- Coronary Artery Disease (CAD)
- Chronic Stable Angina (CSA)
- Acute Coronary Syndrome (ACS)
  - Unstable Angina
  - Myocardial Infarction (MI)

**Review:**

**ATI:**

*Book: Chapter 32; Page 221*

*ATI Online Resources: Engage Adult Medical Surgical RN > Alterations in Health >*

*Alterations in Cardiovascular Function and Perfusion*

*(Sections: A&P of the Cardiac System, Coronary Artery Disease, Heart Failure, Peripheral Vascular Disease, Heart Failure, Peripheral Vascular Disease [aka CVI], Peripheral Artery Disease)*

**Lippincott:**

*Chapter 21: Assessment of Cardiovascular Function*

*Chapter 23: Management of Patients with Coronary Vascular Disorders*

**Lecture Day 8: Perfusion Concepts**

**Concept focus:** blood flow *from the heart* – supporting systemic circulation driven by the heart as a pump through arterial and venous systems

**Objectives:**

By the end of this learning experience, the student will be able to:

1. **Describe** the role of arterial and venous systems in supporting systemic circulation. (QSEN: Patient-Centered Care, Safety)
2. **Identify** assessment findings and diagnostic data associated with impaired peripheral perfusion. (QSEN: Informatics, Safety)
3. **Differentiate** between peripheral artery disease and chronic venous insufficiency based on pathophysiology and clinical presentation. (QSEN: Patient-Centered Care, Evidence-Based Practice, Safety)
4. **Apply** appropriate nursing interventions for patients experiencing alterations in peripheral perfusion. (QSEN: Patient-Centered Care, Teamwork & Collaboration, Safety)
5. **Analyze** patient responses to treatment and management strategies for peripheral vascular disorders to support clinical judgment. (QSEN: Evidence-Based Practice, Safety)

**Exemplars:**

- Peripheral Vascular Disease (PVD)
  - Peripheral Artery Disease (PAD)
  - Chronic Venous Insufficiency (CVI)

**Review:**

**ATI:**

*Book: Chapter 36, Page 249*

*ATI Online Resources: Engage Adult Medical Surgical RN > Alterations in Health > Alterations in Cardiovascular Function and Perfusion*

**Lippincott:**

*Chapter 26 – Assessment and Management of Patients With Vascular Disorders and Problems of Peripheral Circulation*

*\*Chapter 21 – Assessment of Cardiovascular Function*

*\*Chapter 27 – Hypertension*

**Lecture Day 9: Perfusion Concepts**

**Concept focus:** blood flow *through the heart chambers* – supporting the ability to pump blood through systemic circulation.

**Objectives:**

By the end of this learning experience, the student will be able to:

1. **Describe** normal cardiac chamber function and the mechanisms that support effective ventricular filling and ejection. (QSEN: Patient-Centered Care, Safety)
2. **Identify** assessment findings and diagnostic data associated with impaired cardiac pumping and heart failure. (QSEN: Informatics, Safety)
3. **Explain** the pathophysiology and clinical manifestations of congestive heart failure. (QSEN: Patient-Centered Care, Evidence-Based Practice, Safety)
4. **Apply** appropriate nursing interventions for patients experiencing heart failure to support cardiac output and tissue perfusion. (QSEN: Patient-Centered Care, Teamwork & Collaboration, Safety)
5. **Analyze** patient responses to treatment and management of heart failure, including recognition of complications and disease progression. (QSEN: Evidence-Based Practice, Safety)

**Exemplar:**

- Congestive Heart Failure

**Review:**

**ATI:**

*Book: Chapter 33, Page 229*

*ATI Online Resources: Engage Adult Medical Surgical RN > Alterations in Health >*

*Alterations in Cardiovascular Function and Perfusion*

**Lippincott:**

*Chapter 25: Management of Patients with Complications from Heart Disease*

*\*Chapter 21 – Assessment of Cardiovascular Function*

*\*Chapter 22: Management of Patients with Arrhythmias & Conduction Problems (review for ventricular dysrhythmias contributing to HF)*

**Lecture Day 10: Perfusion Concepts**

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**Concept focus:** impaired perfusion due to vascular occlusion

**Objectives:**

By the end of this learning experience, the student will be able to:

1. **Describe** the pathophysiology of venous thromboembolism and its effects on blood flow and tissue perfusion. (QSEN: Patient-Centered Care, Safety)
2. **Identify** assessment findings and diagnostic data associated with deep vein thrombosis and pulmonary embolism. (QSEN: Informatics, Safety)
3. **Differentiate** between the clinical manifestations of DVT and pulmonary embolism. (QSEN: Patient-Centered Care, Evidence-Based Practice, Safety)
4. **Apply** appropriate nursing interventions for patients experiencing venous thromboembolism to prevent complications and promote perfusion. (QSEN: Patient-Centered Care, Teamwork & Collaboration, Safety)
5. **Analyze** patient responses to anticoagulation therapy and other treatments for venous thromboembolism to support clinical judgment. (QSEN: Evidence-Based Practice, Safety)

**Exemplars:**

- Venous Thromboembolism (VTE)
  - Deep Vein Thrombosis (DVT)
  - Pulmonary Embolism (PE)

**Review:**

**ATI:**

*Book: Chapter 25, Page 167*

*ATI Online Resources: Engage Adult Medical Surgical RN > Alterations in Health > Alterations in Hematologic Function > Blood Clots*

**Lippincott:**

*Chapter 26: Assessment and Management of Patients with Vascular Disorders and Problems of Peripheral Circulation*

*\*Chapter 19: Management of Patients with Chest and Lower Respiratory Tract Disorders*

*\*Chapter 28: Assessment of Hematologic Function and Treatment Modalities*

**Lecture Day 11: Infection Concept**

**Concept focus:** alterations in host defense resulting from infectious processes and the body's response to invading pathogens.

**Objectives:**

By the end of this learning experience, the student will be able to:

1. **Describe** the body's normal defense mechanisms and the pathophysiologic response to infection. (QSEN: Patient-Centered Care, Safety)
2. **Identify** assessment findings and diagnostic data associated with common infectious processes. (QSEN: Informatics, Safety)
3. **Differentiate** between respiratory and skin/soft tissue infections based on clinical presentation and transmission patterns. (QSEN: Patient-Centered Care, Evidence-Based Practice, Safety)
4. **Apply** appropriate nursing interventions for patients experiencing infectious processes to prevent complications and limit transmission. (QSEN: Patient-Centered Care, Teamwork & Collaboration, Safety)
5. **Analyze** patient responses to treatment and infection control measures to support clinical judgment and patient safety. (QSEN: Evidence-Based Practice, Safety)

**Exemplars:**

- Pneumonia
- Influenza
- Cellulitis
- Methicillin-Resistant Staphylococcus Aureus (MRSA)

**Review:**

**ATI:**

*Book: Chapter 21 & 75, Pages 143 & 566*

*ATI Online Resources: Engage Adult Medical Surgical RN > Alterations in Health >*

*Alterations in Gas Exchange (**Pneumonia**)*

*Alterations in Immunity & Inflammatory (**Influenza & MRSA/MDRO**)*

*Alterations in Tissue Integrity (**Cellulitis**)*

**Lippincott:**

*Chapter 7: Assessment and Management of Patients with Infectious Diseases*

*Chapter 18: Management of Patients with Upper Respiratory Tract Disorders*

*Chapter 19: Management of Patients with Chest and Lower Respiratory Tract Disorders*

*\*Chapter 13: Skin Integrity and Wound Care*

**Lecture Day 12: Inflammation Concept**

**Concept focus:** inflammatory processes affecting the digestive and gastrointestinal systems.

**Objectives:**

By the end of this learning experience, the student will be able to:

1. **Describe** the inflammatory response and its effects on normal digestive and gastrointestinal function. (QSEN: Patient-Centered Care, Safety)
2. **Identify** assessment findings and diagnostic data associated with inflammatory conditions of the gastrointestinal system. (QSEN: Informatics, Safety)
3. **Differentiate** between GERD, pancreatitis, appendicitis, and hepatitis based on pathophysiology and clinical presentation. (QSEN: Patient-Centered Care, Evidence-Based Practice, Safety)
4. **Apply** appropriate nursing interventions for patients experiencing inflammatory gastrointestinal conditions to support recovery and prevent complications. (QSEN: Patient-Centered Care, Teamwork & Collaboration, Safety)
5. **Analyze** patient responses to treatment and management of gastrointestinal inflammation to support clinical judgment and ongoing care. (QSEN: Evidence-Based Practice, Safety)

**Exemplars:**

- Gastroesophageal Reflux Disease (GERD)
- Pancreatitis
- Appendicitis
- Hepatitis (A, B, C, D, & E)

**Review:**

*ATI Book: Chapter 49, 55, & 56; Page 341, 387, & 395*

*ATI Online Resources: Engage Adult Medical Surgical RN > Alterations in Health > Alterations in Digestion & Bowel Elimination*

**Lippincott:**

*Chapter 38: Assessment of Digestive and Gastrointestinal Function*

*Chapter 39: Management of Patients with Oral and Esophageal Disorders*

*Chapter 41: Management of Patients with Intestinal and Rectal Disorders*

*Chapter 43: Assessment and Management of Patients with Hepatic Disorders*

*\*Chapter 44: Management of Patients with Biliary Disorders*

**EXAM BREAKDOWN BY EXAMPLARS:**

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**EXAM 1 & 3: 75 questions including math calculations.**

**Lecture Day 1: Electrolyte Balance Concepts**

- Sodium Imbalances (Deficit & Excess)
- Potassium Imbalances (Deficit & Excess)
- Calcium Imbalances (Deficit & Excess)
- Magnesium Imbalances (Deficit & Excess)
- Phosphorus Imbalances (Deficit & Excess)
- Chloride Imbalances (Deficit & Excess)

**Lecture Day 2: Acid-Base Balance Concepts**

- Metabolic Acidosis
- Metabolic Alkalosis
- Respiratory Acidosis
- Respiratory Alkalosis

**Lecture Day 3 & 4: Metabolism Concepts**

- Diabetes Type 1
- Diabetes Type 2

**Lecture Day 5: Cellular Regulation Concepts**

- Cellular Regulation
- Iron Deficiency Anemia
- Megaloblastic Anemias
  - Pernicious Anemia
  - Folic Acid Anemia
- Hemolytic Anemia
  - Sickle-Cell Anemia

**Lecture Day 6: Managing Care Concepts**

- Preoperative Care
- Postoperative Care

**EXAM 2 & 4:** 75 questions including math calculations.

**Lecture Day 7: Perfusion Concepts**

- Coronary Artery Disease (CAD)
- Chronic Stable Angina (CSA)
- Acute Coronary Syndrome (ACS)
  - Unstable Angina
  - Myocardial Infarction (MI)

**Lecture Day 8: Perfusion Concepts**

- Peripheral Vascular Disease (PVD)
  - Peripheral Artery Disease (PAD)
  - Chronic Venous Insufficiency (CVI)

**Lecture Day 9: Perfusion Concepts**

- Congestive Heart Failure

**Lecture Day 10: Perfusion Concepts**

- Venous Thromboembolism (VTE)
  - Deep Vein Thrombosis (DVT)
  - Pulmonary Embolism (PE)

**Lecture Day 11: Infection Concept**

- Pneumonia
- Influenza
- Cellulitis
- Methicillin-Resistant Staphylococcus Aureus (MRSA)

**Lecture Day 12: Inflammation Concept**

- Gastroesophageal Reflux Disease (GERD)
- Pancreatitis
- Appendicitis
- Hepatitis (A, B, C, D, & E)

**Final Exam:**

**100 questions – 45 Med-Surg, 45 Maternal-Child, 10 Fundamentals**